Magnetostatics

END EXAM - 2023/24

(03) (a) (i) The line integral $\vec{B}.d\vec{l}$ around any closed path equal to $\mu_0 I$ where I is the total steady current passing through the closed path

$$\oint_l \vec{B}.d\vec{l} = \mu_0 I_{enc}.$$

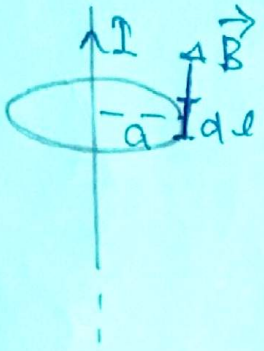
$\vec{B}$: Magnetic field vector

$d\vec{l}$: Infinitesimal vector path of the closed loop

μ_0: permeability of free space

I_{enc}: Total current enclosed by the closed path

(ii)



$$\oint_l \vec{B}.d\vec{l} = \mu_0 I_{enc}.$$

$$B\oint_l dl \cos\theta = \mu_0 I_{enc}$$

$$B(2\pi a).1 = \mu_0 I_{enc}$$

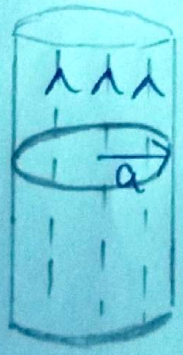
$$B = \frac{\mu_0 I}{2\pi a}$$

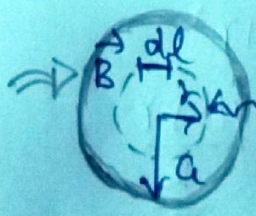
(iii) $I = 5A$, $a = 2\times10^{-2}m$

$$B = \frac{\mu_0 I}{2\pi a} = \frac{(4\pi\times10^{-7} T.m/A)(5.0A)}{2\pi(0.02m)}$$

$$= 5\times10^{-5} T$$

(b) (i)





(1) Inside point ($r < a$)

$$J = \frac{I}{A} = \frac{I}{\pi a^2}$$

$$i = J\pi r^2$$

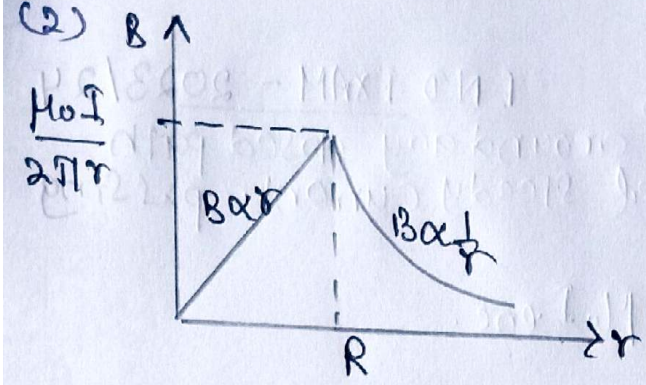
$$\oint_l \vec{B}.d\vec{l} = \mu_0 I_{enc}$$

$$\oint_l B.dl\cos\theta = \mu_0 I_{enc}.$$

$$B(2\pi r) = \mu_0 J\pi r^2 \rightarrow B = \frac{\mu_0 J r}{2}$$

outside point ($r > a$)

$$B(2\pi r) = \mu_0 I \rightarrow B = \frac{\mu_0 I}{2\pi r}$$

(2) 

(ii) $J_{(r)} = J_0\left(1 - \frac{r}{a}\right)$

(1) $I_{enc} = \int J(r)\, dA$

$= \int_0^a J_0\left(1 - \frac{r}{a}\right)(2\pi r)\, dr$

$= 2\pi J_0 \int_0^a \left(r - \frac{r^2}{a}\right) dr$

$I_{enc} = 2\pi J_0 \left(\frac{r^2}{2} - \frac{r^3}{3a}\right)_0^a = 2\pi J_0\left[\left(\frac{a^2}{2} - \frac{a^3}{3a}\right) - 0\right]$

$I_{enc} = 2\pi J_0\left(\frac{a^3}{6}\right) = \frac{\pi J_0 a^2}{3}$

(2) $\oint_l \vec{B}.d\vec{l} = \mu_0 I_{enc}$

Inside $(r<a)$

$B(2\pi r) = \mu_0(2\pi J_0)\left(\frac{r^2}{2} - \frac{r^3}{3a}\right)$

$B = \mu_0 J\left(\frac{r}{2} - \frac{r^2}{3a}\right)$

outside $(r>a)$

$B(2\pi r) = \mu_0 I$

$B(2\pi r) = \mu_0 \frac{\pi J_0 a^2}{3}$ $\Rightarrow B = \frac{\mu_0 J_0 a^2}{6r}$

(3) when J is constant:

Inside $(r<a)$

$I_{enc} = \int_0^r J(2\pi r_1)\, dr_1 = J\pi r^2$

$B(2\pi r) = \mu_0(J\pi r^2) \Rightarrow B = \frac{\mu_0 J r}{2}$

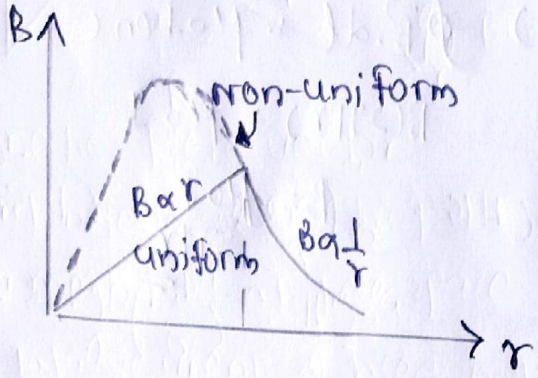
outside $(r>a)$

$I_{enc} = \int_0^a J(2\pi r)\, dr = J\pi a^2$

$B(2\pi r) = \mu_0 J\pi a^2$

$B = \frac{\mu_0 J a^2}{2r}$

In the uniform case: $\vec{B}$ insider the conductor (Bar) in the non-uniform case $J = J_0(1 - r/a)$ as r increases additional current is enclosed per unit step compared to the uniform model. So, B grow more slowly than linear.



END EXAM - 2022/23

(Q2) (a) (i) The line integral $\vec{B} \cdot d\vec{l}$ around any closed path equal to $\mu_0 I$ where I is the total steady current passing through the closed path

$$\oint_l \vec{B}\, d\vec{l} = \mu_0 I_{enc}$$

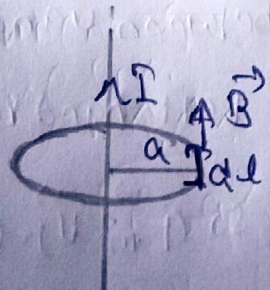
$\vec{B}$: Magnetic field vector

$d\vec{l}$: Infinisimal vector path element of the closed path

μ_0: permeability of free space

I_{enc}: Total current enclosed by the closed path

(ii)

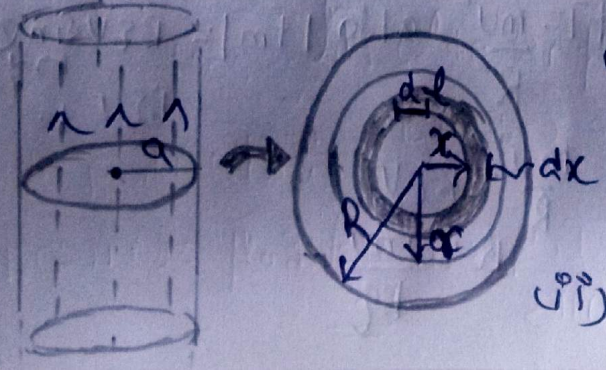


$$\oint_l \vec{B} \cdot d\vec{l} = \mu_0 I_{enc} \Rightarrow \oint_l B \cdot dl \cos 0 = \mu_0 I_{enc}$$

$$B(2\pi a) = \mu_0 I$$

$$B = \frac{\mu_0 I}{2\pi a}$$

(b)



(i) $I_{enc} = \int di = \int_0^r J\, dA$

$$= \int_0^r (Kx)(2\pi x)\, dx = \int_0^r 2\pi K x^2\, dx$$

(ii) $I_{enc} = 2\pi K \left[\frac{x^3}{3}\right]_0^r = \frac{2}{3}\pi K r^3$

(Q2) (i) $I_T = I_{enc}(R)$ $\int D$ $K = \frac{3I}{2\pi R^3}$

$I = \frac{2}{3}\pi K R^3$

(ii) $\oint \vec{B}.d\vec{l} = \mu_0 I_{enc}$.

$B(2\pi r) = \mu_0 (\frac{2}{3}\pi K r^3)$

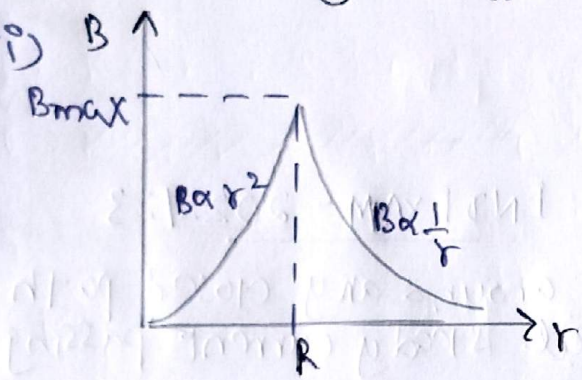
$B = \frac{1}{3}\mu_0 r^2 K$

(v) $\oint \vec{B}.d\vec{l} = \mu_0 I_{enc}$

$B(2\pi r) = \mu_0 I$

$B = \frac{\mu_0 I}{2\pi r}$

(vi)



(Q3) (a) (i) combined force exerted on a moving charged particle by electric & magnetic fields.

$\vec{F_L} = q\vec{E} + q(\vec{v} \times \vec{B})$

Elect. Mgn.

mg component:

$\vec{f_m} = q(\vec{v} \times \vec{B})$

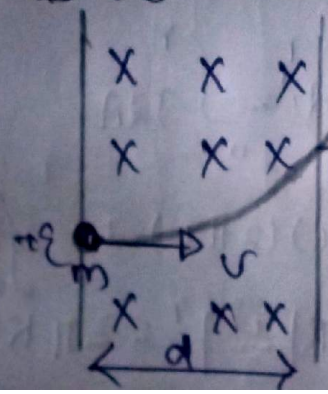
$|f_m| = q(Bqv\sin\theta)$ where θ is the $\angle$ between $\vec{v}$ and $\vec{B}$

direction: $\perp$ to both $\vec{v}$ and $\vec{B}$, using Fleming's Left hand rule.

(ii) $W = \int \vec{F}.d\vec{s}$, In $\vec{f_m} = q(\vec{v} \times \vec{B})$ $\vec{f_m} \perp \vec{v}$ so $(\vec{f_m}.\vec{v}) = 0$

So $\vec{F}.d\vec{s} = 0$

(b)



(i) $|\vec{F}| = \frac{mv^2}{R}$ also $|\vec{f_m}| = Bqv\sin\theta$

$= Bqv\,1.\ (\theta = \pi/2)$

$\frac{mv^2}{R} = Bqv$

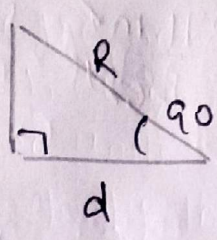
$R = \frac{mv}{Bq}$

$t = \left(\frac{\theta}{2\pi}\right) T \quad \therefore \quad T = \frac{2\pi R}{v}$

$T = \frac{2\pi}{v}\left(\frac{mv}{Bq}\right)$

$T = \frac{2\pi m}{Bq}$

$d = \frac{mv}{2qB} \qquad d = \frac{R}{2}$



$\sin\theta = \frac{d}{R} = \frac{R}{2R}$

$R\sin\theta = d$

$\theta = 30°$

$t = \left(\frac{\pi/6}{2\pi}\right)\frac{2\pi m}{Bq} = \frac{m\pi}{6Bq}$

$\therefore t = \frac{m\pi}{6Rq}$

(2) $d = \frac{mv}{qB} \qquad d = R$

$\sin\theta = 1$

$\theta = 90°$

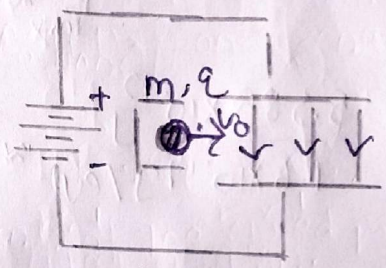
$t = \left(\frac{\pi/2}{2\pi}\right)\left(\frac{2\pi m}{Bq}\right) = \frac{m\pi}{2Bq}$

(3) $d = \frac{2mv}{qB} \qquad d = 2R$

$\theta = 180°$

$t = \left(\frac{\pi}{2\pi}\right)\left(\frac{2\pi m}{Bq}\right) = \frac{m\pi}{Bq}$

(c) (i) $\vec{E} \parallel \vec{B}$

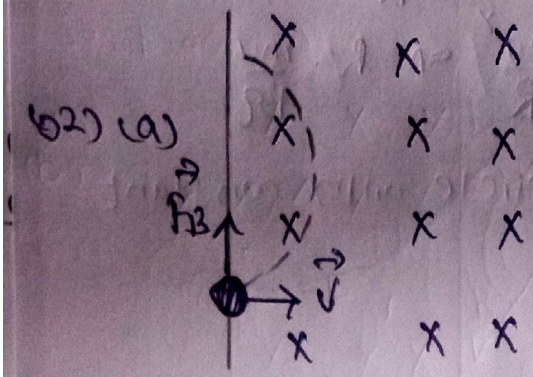


- perpendicular velocity component interacts with $\vec{B}$ to create uniform circular motion
- parallel velocity comp. interacts with $\vec{E}$ to create linear acceleration.

The superposition of these results a helical path with an increasing pitch.

(ii) $F_E = qE \qquad F_B = Bqv$

which results: $v = \frac{E}{B}$ so particle moves in a straight line.



END EXAM - 2019/20

(b2) (a)

$f_{mag} = F_{centripetal}$

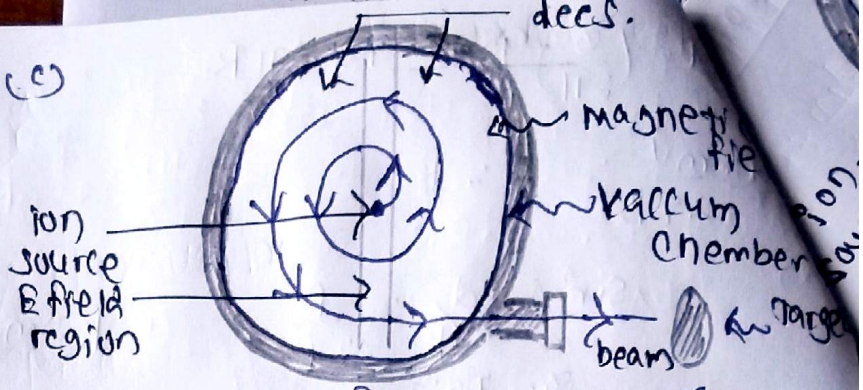
$qvB\sin 90° = \frac{mv^2}{R}$

$Bqv = \frac{mv^2}{R} \Rightarrow R = \frac{mv}{Bq}$

(b) $T = \frac{2\pi R}{v} = \frac{2\pi mv}{BQv}$

$T = \frac{2\pi m}{BQ}$

(c)



(d) $r = 8.0 \times 10^{-2}\,m \quad B = 1.5\,T \quad m = 3.3 \times 10^{-27}\,kg \quad Q = 1.6 \times 10^{-19}\,C$

$f = \frac{BQ}{2\pi m} = \frac{(1.5)(1.6 \times 10^{-19}\,C)}{2\pi(3.3 \times 10^{-27})} \approx 1.157 \times 10^{7}\,Hz$

(e) $E = \frac{1}{2}mv^2$

$BQv = \frac{mv^2}{R} \Rightarrow \frac{BQR}{m} = v \Rightarrow E = \frac{1}{2}m\left(\frac{BQR}{m}\right)^2$

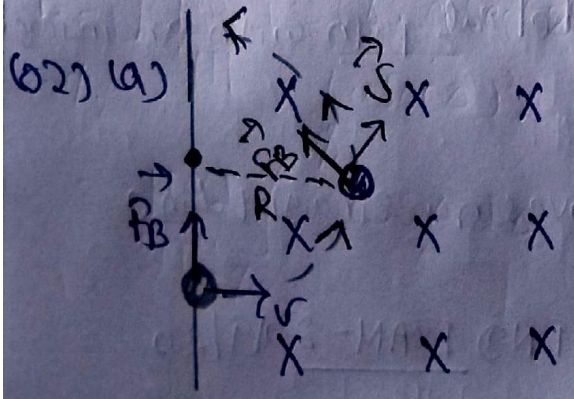
$E = \frac{B^2Q^2R^2}{2m} = \frac{(1.6 \times 10^{-19})^2(1.5)^2(0.8)^2}{2 \times 3.3 \times 10^{-27}}$

$E_{max} \approx 5.585 \times 10^{-12}\,J$

$E_{max}\,(in\ eV) = \frac{5.585 \times 10^{-12}\,J}{1.6 \times 10^{-19}\,J/eV} \approx 3.49 \times 10^{7}\,eV$

END EXAM-2016/2017

(02) (a)

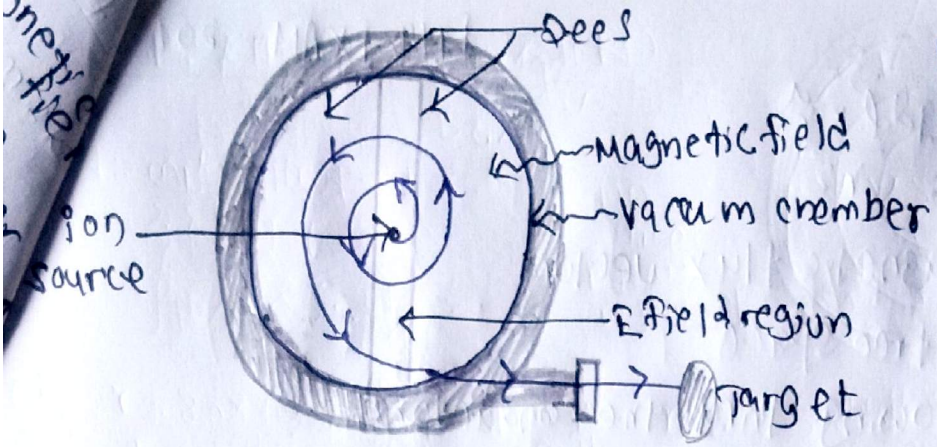


$F_{mag} = F_{centripetal}$

$BQv\sin\theta = \frac{mv^2}{R}$

$BQv = \frac{mv^2}{R} \Rightarrow R = \frac{mv}{BQ}$

The path of the charged particle is a circle with constant radius of $R = mv/BQ$



(i) $T = \frac{2\pi R}{v}$

$T = \frac{2\pi}{v}\frac{mv}{Bq}$

$T = \frac{2\pi m}{Bq}$

$f = \frac{Bq}{2\pi m}$

(ii) $R = 0.5\,m \quad B = 0.25\,T \quad m = 6.65 \times 10^{-27}\,kg$

α-particle $(He^{2+}) = 2e = 2 \times 1.6 \times 10^{-19}\,C$

$$f = \frac{(0.25)(3.2 \times 10^{-19})}{2\pi(6.65 \times 10^{-27})} \approx 1.914 \times 10^{6}\,Hz$$

(iii) $KE_{max} = \frac{1}{2}mv^2 \quad \because Bqv = \frac{mv^2}{R} \Rightarrow \frac{BqR}{m} = v$

$= \frac{1}{2}m\frac{B^2q^2R^2}{m^2}$

$$KE_{max} = \frac{B^2q^2R^2}{2m} = \frac{(0.25)^2(3.2 \times 10^{-19})^2(0.5)^2}{2(6.65 \times 10^{-27})}$$

$= 1.203 \times 10^{-13}\,J$

$$KE_{max}\,(in\ eVs) = \frac{1.203 \times 10^{-13}\,J}{1.6 \times 10^{-13}\,J/MeV} \approx 0.752\,MeV$$

(iv) Relativistic Effects: As particles gain immense KE and approach the speed of light (c), their relativistic mass increases ($\downarrow f \propto \frac{1}{m} \downarrow$ f decrease. The particles fall out of phase with the constant frequency of the altering accelerating voltage.

Size and cost of Electromagnets: $KE_{max} \propto R^2$, so reaching extremely high energies requires an extra large radius. Building solid iron core electromagnets large enough to generate $\vec{B}$ is financially infeasible.

END EXAM - 20

(01) (a) $d\vec{B} = \left(\frac{\mu_0}{4\pi}\right) \frac{I d\vec{l} \times \hat{r}}{r^2}$

$d\vec{B}$: Infinitisimal magnetic flux vector

μ_0: Permiability of free space

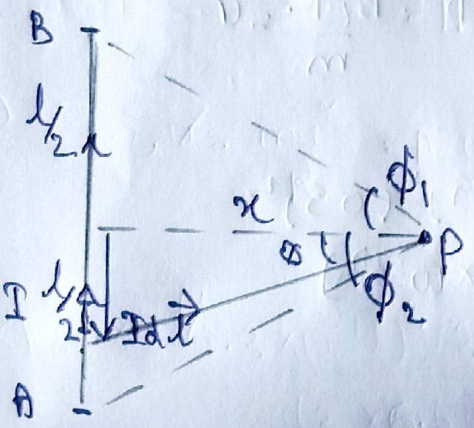
I: Steady current flowing thru the conductor

$d\vec{l}$: Infinitisimal vector length elemt of the conductor pointing the direction of the current

$\vec{r}$: position vector directed from the $d\vec{l}$

$\hat{r}$: ~~Magnitude of r~~ unit vector in the direction of $\vec{r}$

(b)

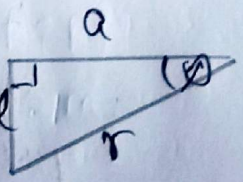


$$d\vec{B} = \left(\frac{\mu_0}{4\pi}\right)\left(\frac{I d\vec{l} \times \hat{r}}{r^2}\right)$$

$$= \left(\frac{\mu_0}{4\pi}\right) \frac{I|d\vec{l}||\hat{r}|\sin\theta}{r^2}$$

$$= \frac{\mu_0}{4\pi} \frac{I\cos\theta \cdot dl}{r^2}$$

$$B = \int_A^B \frac{\mu_0}{4\pi} \frac{I\cos\theta\, dl}{r^2}$$



$(r = a\sec\theta)$

$(l = a\tan\theta)$

$dl = a\sec^2\theta\, d\theta$

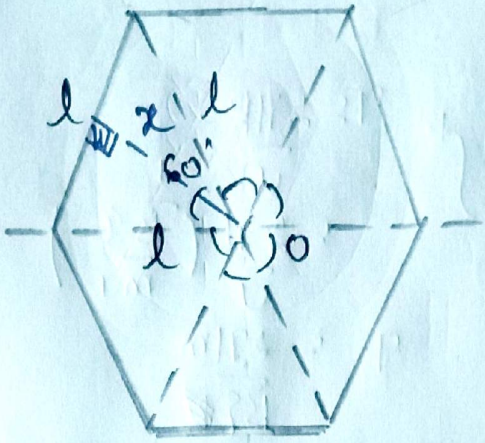
$$B = \int_{-\phi_2}^{\phi_1} \frac{\mu_0}{4\pi} \frac{I\cos\theta \cdot a\sec^2\theta\, d\theta}{(a\sec\theta)^2}$$

$$= \frac{\mu_0 I}{4\pi a} \int_{-\phi_2}^{\phi_1} \cos\theta\, d\theta \Rightarrow B = \frac{\mu_0 I}{4\pi a} \sin\theta \Big|_{-\phi_2}^{\phi_1}$$

$$B = \frac{\mu_0 I}{4\pi a} (\sin\phi_1 + \sin\phi_2)$$

Here $B = \frac{\mu_0 I}{4\pi a} 2\sin\phi \quad (a = x)$

$$B = \frac{\mu_0 I (2\sin\phi)}{4\pi x}$$



$x = l\cos 30° = \frac{l\sqrt{3}}{2}$

$x = \frac{\sqrt{3}l}{2}$

now $B = \frac{\mu_0 I}{4\pi x}(2\sin\phi)$

$= \frac{\mu_0 I}{4\pi\left(\frac{\sqrt{3}l}{2}\right)} \cdot (2\sin 30°)$

$B = \frac{\mu_0 I}{2\pi\sqrt{3}l}$

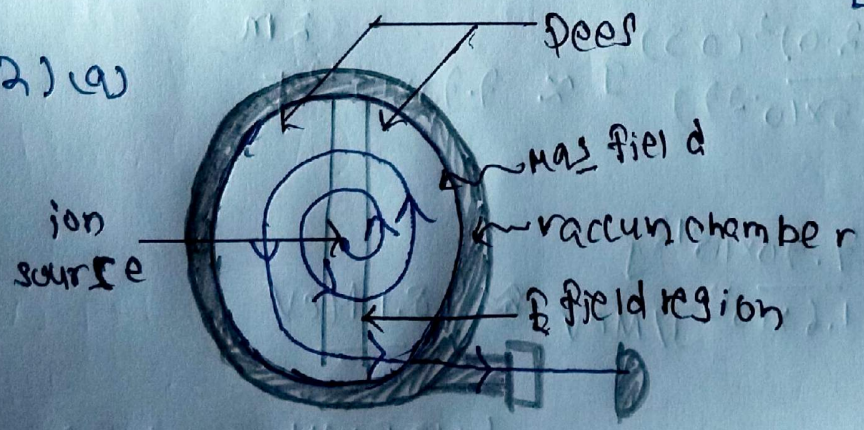
for all the wires in the Hexagon:

$B_T = 6 \times N\left(\frac{\mu_0 I}{2\pi\sqrt{3}l}\right) = N\frac{\sqrt{3}\mu_0 I}{\pi l}$

$B_T = \frac{50 \times \sqrt{3} \times 4\pi \times 10^{-7} \times 2\,A}{\pi \times 0.02} \approx 3.46 \times 10^{-3}\,T$

END EXAM - 2014/15

(02) (a)



A cyclotron accelerates charged particles using a comb'n of $\vec{E}$ and $\vec{B}$. consists of 02 hollow semi-circular metal electrodes (dees) in a vaccum chamber. $\vec{B}$ is $\perp^{to}$ these dees. Altern ative high frequency voltage is applied across the narrow gap seperat -ing them. when an ion is injected at the center. $\vec{B}$ exerts a force $\perp$ to particle velocity, causing it to move semicircular path. Every time particle crosses the gap $\vec{E}$ accelerates it. $r \uparrow$ $v \uparrow$ so orbital radius $\uparrow$ (r↑v) This continues until particle goes ing

spiral and exits as a high-energy beam.

(b) $F_{mag} = F_{centripetal}$

$$Bqv\sin 90^\circ = \frac{mv^2}{R} \Rightarrow v/R = \frac{Bq}{mv}$$

$$Bqv = \frac{mv^2}{R}$$

$$T = \frac{2\pi R}{v} = 2\pi\left(\frac{1}{v}\right)\left(\frac{Bq}{mv}\right)$$

$$T = \frac{2\pi m}{Bq}$$

So $f = \frac{1}{T} \Rightarrow f = \frac{Bq}{2\pi m}$

(c) $R = 50\times10^{-2}\,m$, $B = 2.0\,T$, $q = 3.2\times10^{-19}$, $m = 6.65\times10^{-27}$

$$f = \frac{(3.2\times10^{-19})(2.0)}{2\pi(6.65\times10^{-27})} \simeq 1.53\times10^{7}\,Hz$$

(d) $Bqv = \frac{mv^2}{R} \Rightarrow v = \frac{Bqr}{m} \Rightarrow KE_{max} = \frac{1}{2}m\left(\frac{BqR}{m}\right)^2$

$$= \frac{(BqR)^2}{2m}$$

$$KE_{max} = \frac{(3.2\times10^{-19})^2(2.0)^2(0.5)^2}{(2\times6.65\times10^{-27})} J \simeq 7.70\times10^{-12}\,J$$

$$KE_{max}\,(in\ MeV) = \frac{7.70\times10^{-12}\,J}{1.6\times10^{-13}\,J/MeV} \approx 48.1\,MeV$$

END EXAM - 2012/2013

(01) (a) $\oint_l \vec{B}\cdot d\vec{l} = \mu_0 I_{enclosed}$

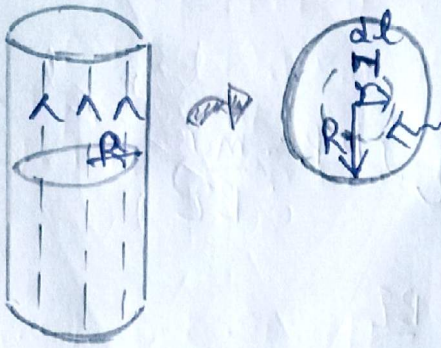
The line integral $\vec{B}\cdot d\vec{l}$ around any closed path equals to $\mu_0 I$ where I is the total current passing through the closed path.

$\vec{B}$: Magnetic field vector

$d\vec{l}$: Infinitisimal vector path of the closed loop

μ_0: Permeability of free space. I_{enc}: Total steady current passing

Outside point ($r \geq R$)



$$\oint_l \vec{B}\cdot d\vec{l} = \mu_0 I_{enc}$$

$$\oint_l B\, dl \cos\theta = \mu_0 I$$

$$B(2\pi r) = \mu_0 I$$

$$B = \frac{\mu_0 I}{2\pi r}$$

Inside point ($r \leq R$)

$$J = \frac{I}{\pi R^2}$$

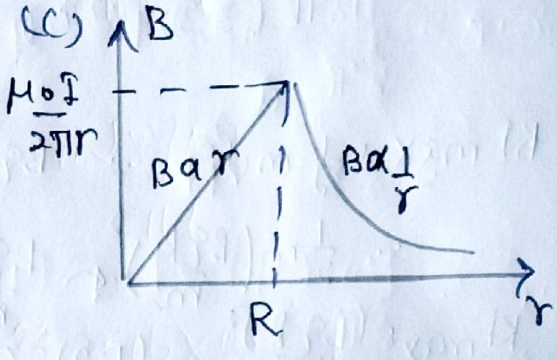
$$I_{enc} = J(\pi r^2)$$

$$I_{enc} = \frac{I}{\pi R^2}\pi r^2 = \frac{I r^2}{R^2}$$

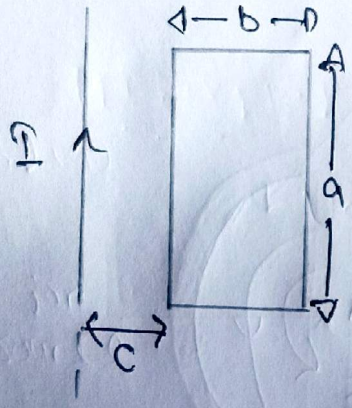
$$\oint_l \vec{B}\cdot d\vec{l} = \mu_0 I_{enc}$$

$$B(2\pi r) = \mu_0 \frac{I r^2}{R^2} \rightarrow B = \frac{\mu_0 I r}{2\pi R^2}$$

(c)



(d)



$$B(x) = \frac{\mu_0 I}{2\pi x} \quad \therefore d\Phi = B(x)\,dA$$

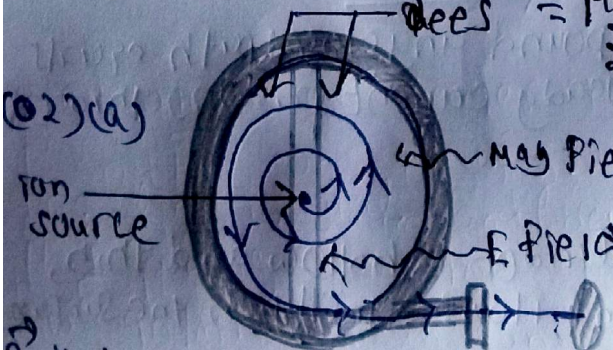
$$= \frac{\mu_0 I}{2\pi x} a\,dx$$

$$= \frac{\mu_0 I a\,dx}{2\pi x}$$

$$\Phi = \int_c^{c+b} \frac{\mu_0 I a}{2\pi x} dx = \frac{\mu_0 I a}{2\pi}\int_c^{c+b} \frac{1}{x} dx$$

$$= \frac{\mu_0 I a}{2\pi} \ln\left(\frac{c+b}{c}\right)$$

(02)(a)



A cyclotron accelerates charged particles using a combn of $\vec{B}$ & $\vec{E}$ consists of 02 hollow metal ~~conductors~~ electrodes (dees) in a vaccum chamber.

$\vec{B}$ is ⊥ to these dees. Alternative high freq voltage is applied across the narrow gap seperating them. when an ion is injected at the center $\vec{B}$ exerts a force ⊥ to particle V, causing to move semicircular path Every time it crosses $\vec{E}$ gap $\vec{E}$ accelerates it $K = \frac{1}{2}mv^2 = qV$

This continues until it exits as a high energy beam.

(b) $F_{centripetal} = F_{mag}$

$\frac{mv^2}{R} = Bqv \Rightarrow \frac{mv}{R} = Bq$ $\quad R = \frac{mv}{Bq}$

$T = \frac{2\pi R}{v} = \frac{2\pi}{v}\left(\frac{mv}{Bq}\right) = \frac{2\pi m}{Bq} \Rightarrow f = \frac{Bq}{2\pi m}$

(c) $f = \frac{1.5 \times 1.6\times10^{-19}}{2\pi\,(3.3\times10^{-27})} \approx 1.157 \times 10^{7}\ Hz$

(d) $KE_{max} = \frac{1}{2}mv^2 \quad \because Bqv = \frac{mv^2}{R}$

$= \frac{1}{2}m\left(\frac{BqR}{m}\right)^2 \quad \because \frac{Bq}{m} = f \Rightarrow v = fR$

$KE_{max} = \frac{(BqR)^2}{2m} = \frac{(1.5)^2(1.6\times10^{-19})^2(80\times10^{-2})^2}{2\times3.3\times10^{-27}}$

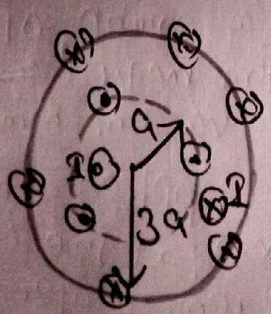
$KE_{max} \approx 5.585 \times 10^{-12}\ J$

$KE_{max}\ (in\ MeV) = \frac{5.585\times10^{-12}}{1.6\times10^{-13}\ J/MeV} \approx 34.9\ MeV$

END EXAM - 2010

(02) (a) The line integral $\vec{B}\cdot d\vec{l}$ around any closed path equals to $\mu_0 I$, where I is the total steady current passing through to closed path.

(b)



(i) $r < a$

Since the inner cylinder is hollow and thin, all of its current flows strictly on the surface $r = a$ so $I_{enc} = 0$

$\oint_L \vec{B}\,d\vec{l} = \mu_0 \cdot 0 \Rightarrow B(2\pi r) = \mu_0 = 0$

$a < r < 3a$

$I_{enc} = I$ ∫P $\oint_l \vec{B}.d\vec{l} = \mu_0 I_{enc}$ ∫P $B = \frac{\mu_0 I}{2\pi r}$ counter-clock wise

$B(2\pi r) = \mu_0 I$

(ii) $r > 3a$

Now $I_{enc} = I_{in} - I_{outer} = I - 3I = -2I$ ⊙ $= 2I$ ⊗

$B(2\pi r) = \mu_0 |-2I|$

$B = \frac{\mu_0 I}{\pi r}$ ↻ clock-wise.

END EXAM - 2021/2022

(03)(a) $d\vec{B} = \left(\frac{\mu_0}{4\pi}\right) \cdot \frac{I d\vec{l} \times \hat{r}}{r^2}$

$d\vec{B}$: Infinitisimal magnetic flux vector

$d\vec{l}$: " vector length element

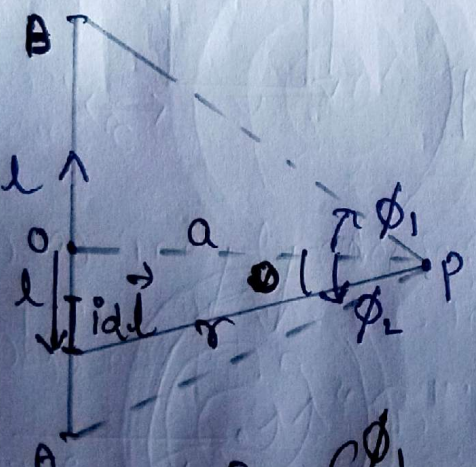
I : Steady current flowing thru the conductor

μ_0 : permeability of free space.

r : position vector directed from the $d\vec{l}$

$\hat{r}$: unit vector in the direction of r

(b)



$d\vec{B} = \left(\frac{\mu_0}{4\pi}\right) \frac{I d\vec{l} \times \hat{r}}{r^2} = \left(\frac{\mu_0}{4\pi}\right) \frac{I |d\vec{l}| |\hat{r}| \sin\alpha}{r^2}$

$= \frac{\mu_0}{4\pi} \frac{I dl \cos\theta}{r^2}$ (where $\cos\theta = \frac{a}{r}$)

$B = \int_A^B \frac{\mu_0}{4\pi} \frac{I \cos\theta dl}{r^2} = \frac{\mu_0 I}{4\pi} \int_A^B \frac{\cos\theta dl}{r^2}$

Also $l = a\tan\theta$ $r = a\sec\theta$

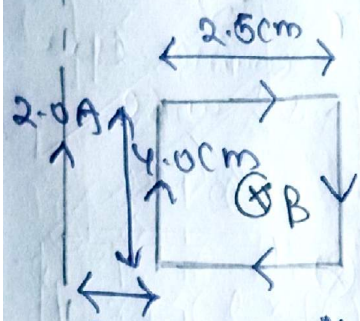
$dl = a\sec^2\theta d\theta$

$B = \int_{-\phi_2}^{\phi_1} \frac{\mu_0}{4\pi} \frac{I\, a\sec^2\theta d\theta \cos\theta}{(a^2\sec^2\theta)} = \frac{\mu_0 I}{4\pi a} (\sin\phi_1 + \sin\phi_2)$

for an infinitely long wire: $\phi_1 = \pi/2$ $\phi_2 = \pi/2$

$B = \frac{\mu_0 I}{4\pi a} (\sin\pi/2 + \sin\pi/2) = \frac{\mu_0 I}{2\pi a}$ //

(c) (i) All the magnetic field lines produced by the long wire are ⊥ to the plane of loop - ⊗ ; As a result All magnetic forces on 04 segments lie entirely within the plane of loop. so there is no tendency to rotate.



$\tau = 0$

(ii) current (i) flows in opposite directions in top and bottom so

$$F_{top} + F_{bottom} = 0$$

$$F_{near} = \frac{\mu_0}{2\pi}\frac{(I_1 I_2)}{a} L = \frac{(4\pi \times 10^{-7})(2.0)(1.0)(0.04)}{2\pi(0.02)}$$

$$= 8 \times 10^{-7} N$$

$$F_{far} = \frac{\mu_0}{2\pi}\frac{(I_1 I_2)}{a} L = \frac{(4\pi \times 10^{-7})(2.0)(1.0)(0.04)}{2\pi(2 + 2.5 \times 10^{-2})}$$

$$\approx 3.56 \times 10^{-7} N$$

$$F_{net} = F_{near} - F_{far} = (8.00 - 3.56) \times 10^{-7} = 4.44 \times 10^{-7} N$$